1.INTRODUCTION

Traditional lock systems using mechanical lock and key mechanism are being replaced by new advanced techniques of locking system. These techniques are an integration of mechanical and electronic devices and highly intelligent. One of the prominent features of these innovative lock systems is their simplicity and high efficiency. Such an automatic lock system consists of electronic control assembly which controls the output load through a password. This output load can be a motor or a lamp or any other mechanical/electrical load. Here we develop an electronic code lock system using 8051 microcontrollers, which provides control to the actuating the load. It is a simple embedded system with input from the keyboard and the output being actuated accordingly. This system demonstrates a password-based door lock system where in once the correct code or password is entered, the door is opened and the concerned person is allowed access to the secured area. Again, if another person arrives it will ask to enter the password. I the password is wrong then door would remain closed, denying the access to the person.

"Password Based Door Security System using Microcontroller" is used in the places where we need more security. It can also use to secure lockers and other protective doors.

The system comprises a number keypad and the keypads are connected to the 8-bit microcontroller AT89C51. This is one of the popular Microcontrollers. It has only 40 pins and there are 32 input/output lines. The microcontroller has a program memory of 2Kilobytes. The microcontroller continuously monitors the keypad and if somebody enters the password it will check the entered password with the password which was stored in the memory and if it they are same then the microcontroller will switch on the corresponding device.

The system will allow the person who knows the password and it will not allow who don't know the password and the system will also show the persons who try to break the protection barrier.

1.1 Need

A password-based door lock system is needed to provide a more secure and reliable alternative to traditional key-based locks. In conventional systems, keys can be easily lost, stolen, or duplicated, which increases the risk of unauthorized access. By using a password system controlled by a microcontroller such as the AT89C51 microcontroller, access is granted only to users who know the correct password, thereby improving security. This system also offers greater convenience because there is no need to carry physical keys, and the password can be changed whenever required. Additionally, features like buzzer alerts for wrong password attempts enhance safety. Due to its low cost, ease of use, and ability to integrate with modern electronic systems, a password-based door lock system is widely used in homes, offices, and security applications, making it an efficient solution for access control.

A password-based door lock system is needed to enhance security, convenience, and reliability in modern access control systems. Traditional locking mechanisms depend on physical keys, which can be easily lost, stolen, or duplicated, making them less secure. In contrast, a password-based system controlled by a microcontroller such as the AT89C51 microcontroller ensures that only authorized users can access the door by entering the correct password. This significantly reduces the chances of unauthorized entry. Additionally, it eliminates the need to carry keys, making it more convenient for users in daily life.

Another important need for this system is its flexibility. The password can be easily changed whenever required, which is not possible with traditional locks unless the entire lock is replaced. It can also be programmed to include security features such as limiting the number of wrong attempts, activating a buzzer alarm, or temporarily locking the system after repeated incorrect entries. These features provide an extra layer of protection.

Moreover, such systems are cost-effective and can be easily implemented using basic electronic components, making them suitable for homes, offices, lockers, and industrial applications. They also serve as an important learning tool in the field of embedded systems, helping students understand microcontroller programming, interfacing, and real-time applications. Overall, the need for a password-based door lock system arises from the growing demand for secure, efficient, and user-friendly access control solutions in today's world.

1.2 Objective

Objectives of the Project

- **To design and implement a password-based door lock system using the AT89C51 microcontroller: -** The primary objective of this project is to create a complete electronic door locking system that operates using a password. The microcontroller is used to control all system operations such as reading input, processing data, and controlling output devices, making the system intelligent and automated.

- **To enhance security by replacing traditional key-based systems: -**This project aims to overcome the limitations of conventional locks, where keys can be lost, stolen, or duplicated. By using a password-based mechanism, access is restricted only to authorized users, thereby improving overall security.

- **To interface input and output devices effectively: -**The system integrates multiple hardware components such as a keypad for input, an LCD for display, and a motor driver like the L293D motor driver IC for controlling the door mechanism. The objective is to understand and implement proper interfacing between these devices and the microcontroller.

- **To control door operation using a motor driver: -**Since the microcontroller cannot drive a motor directly, the project uses a motor driver IC to control the opening and closing of the door. This objective focuses on learning how to control mechanical movement through electronic circuits.

- **To develop embedded programming skills: -**Another key objective is to write and implement embedded C programs that define the working logic of the system. This includes password checking, condition handling, and controlling outputs based on user input.

- **To provide user-friendly interaction through LCD display: -**The system is designed to communicate with the user by displaying messages such as instructions, status updates, and error messages on an LCD screen, making the system easy to use.

- **To implement security features for incorrect password attempts:** -The project also aims to include safety features such as activating a buzzer or denying access when an incorrect password is entered, ensuring protection against unauthorized access

- **To understand real-time application of embedded systems:** -This project helps in understanding how embedded systems are used in real-life applications by combining hardware and software to solve practical problems.

- **To design a cost-effective and efficient system:** -The objective is to build a system that is affordable, reliable, and suitable for use in homes, offices, and other security applications.

1.3 Proposed Work

The proposed work of this project is to design and implement a secure and efficient password-based door lock system using the AT89C51 microcontroller. The system is intended to replace traditional key-based locking mechanisms by introducing a digital access control method that operates using a user-defined password.

In this system, a keypad is used to enter the password, which is then read and processed by the microcontroller. The entered password is compared with a predefined password stored in the microcontroller's memory. If the password matches, the microcontroller sends a control signal to the L293D motor driver IC, which in turn drives a motor to open the door. At the same time, the LCD display shows a message such as "Correct" to inform the user.

If the entered password is incorrect, the system denies access and activates a buzzer to alert about the unauthorized attempt. The LCD displays "Incorrect" or "Access Denied." The system may also be programmed to restrict access after multiple incorrect attempts, thereby enhancing security.

The proposed work also includes proper interfacing of all hardware components such as the keypad, LCD, motor driver, and buzzer with the microcontroller. Supporting components like capacitors and a crystal oscillator are used to ensure stable operation and accurate timing of the system.

Overall, the project focuses on creating a reliable, low-cost, and user-friendly security system that demonstrates the practical application of embedded systems in real-world scenarios.

2. SYSTEM DESIGN

2.1 Block Diagram

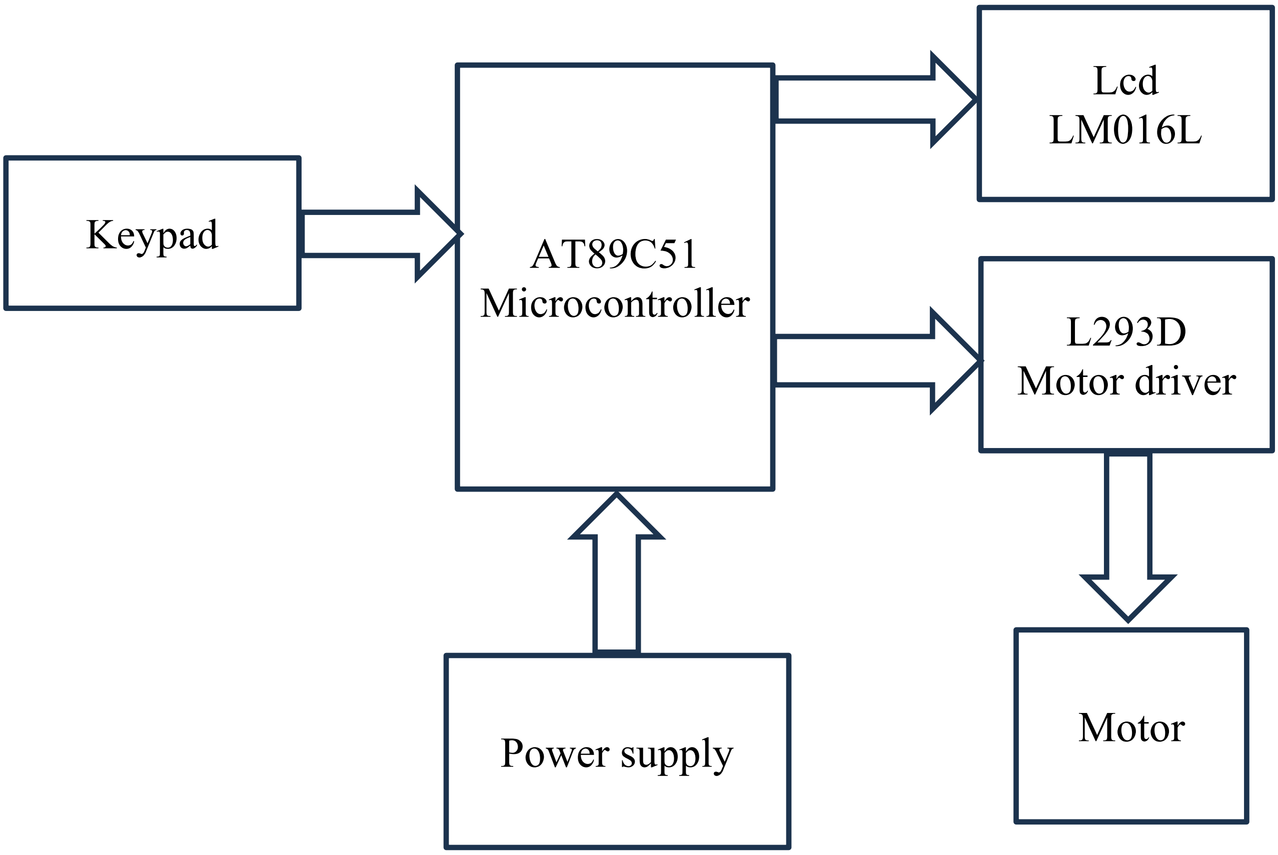


Fig2.1 Block Diagram of Password Based Door Lock System

The block diagram of the password-based door lock system shows how different components are connected and how the system works. The main components include a keypad, microcontroller, LCD display, motor driver, and motor.

Thus, the block diagram represents the flow of data from the keypad (input) to the microcontroller (processing) and then to the LCD and motor (output), ensuring proper and secure operation of the door lock system.

2.2 Circuit Simulation

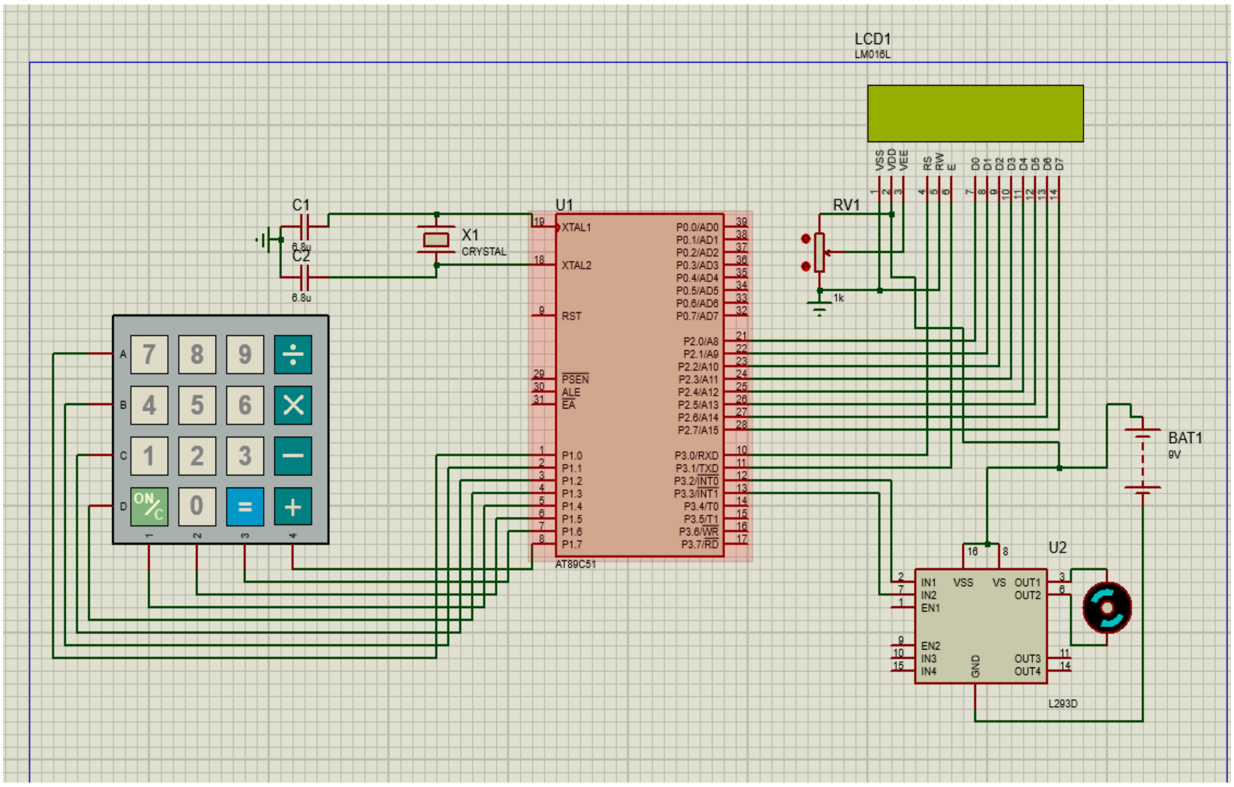


Fig. 2.2 Circuit Simulation of Password Based Door Lock System

The above circuit diagram is password-based door lock system. The circuit is built around the AT89C51 microcontroller, which controls the entire system. The keypad is connected to Port 1 and is used to enter the password. The microcontroller reads this input and processes it. The LCD display is connected to Port 2 and shows messages like password entry and access status, while a potentiometer is used to adjust its contrast.

A crystal oscillator with capacitors is connected to the microcontroller to provide the required clock signal for proper operation. For door control, the microcontroller is interfaced with the L293D motor driver IC, which drives the motor. The motor operates the door lock mechanism by opening or closing it based on the password. The entire circuit is powered by a DC supply, ensuring smooth and reliable functioning of the system.

2.3 PCB Layout and Hardware

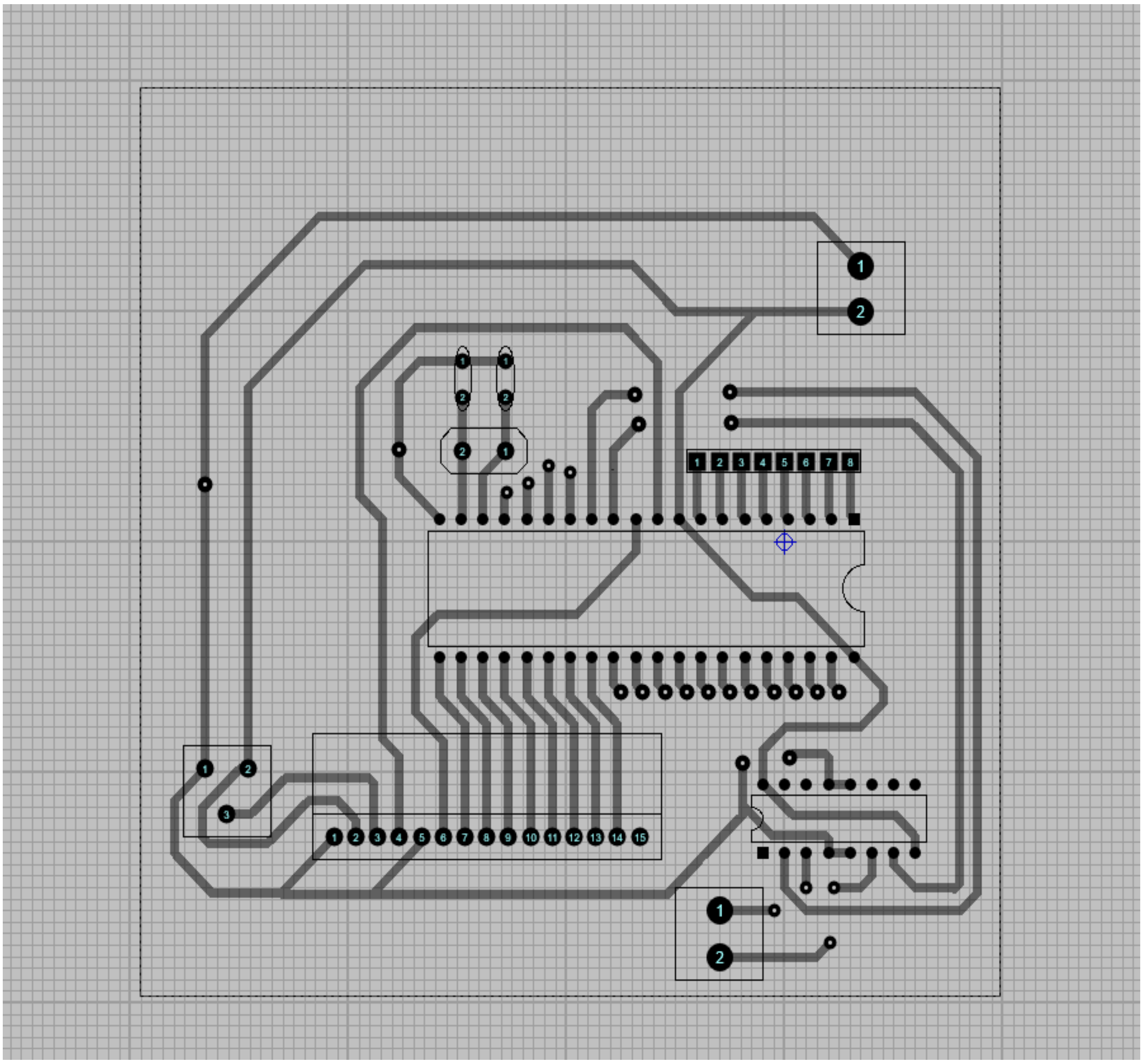


Fig.2.3 PCB Layout

The PCB layout represents the physical design of the circuit where all components are connected through copper tracks. The AT89C51 microcontroller is placed at the center, acting as the main controller. It is connected to the keypad for input and the LCD for displaying messages through organized track lines.

The L293D motor driver IC is connected to the microcontroller to drive the motor, which controls the door mechanism. The layout also includes sections for power supply, ground connections, and crystal oscillator for stable operation.

Overall, the design ensures proper signal flow, compact arrangement, and easy implementation of the password-based door lock system.

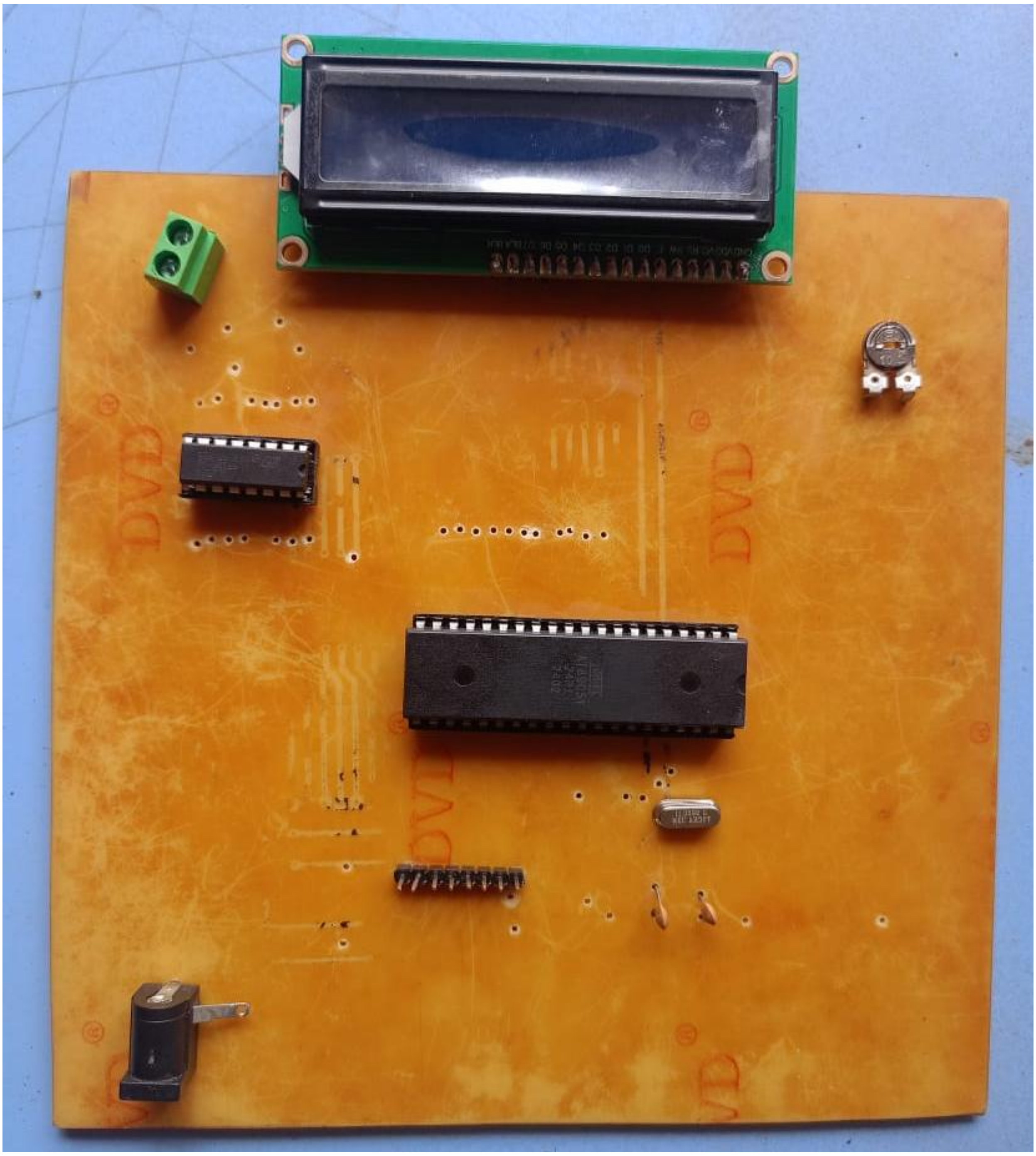


Fig.2.4 Hardware

2.4 Component Description

1. Power Supply (9V Battery)

- **Function:-**

Processes input signals and controls the overall operation of the system.

- **Role-in-Circuit:-**

The battery in this project serves as the main power source that supplies the required DC voltage to the 8051 microcontroller and all connected components, ensuring continuous operation of the traffic light system.

Fig.2.5 Power Supply (9V Battery)

2.Capicator (33pf)

- **Function: -**

A 33pF capacitor is a small-value capacitor mainly used in high-frequency electronic circuits, especially in combination with a crystal oscillator. Its most important function is to provide the required load capacitance that allows the crystal to start and maintain stable oscillations. In microcontroller circuits such as the AT89C51, two 33pF capacitors are typically connected from each crystal pin to ground. These capacitors form part of the oscillator network and help the crystal generate a precise and stable clock signal, which is essential for correct timing, instruction execution, and synchronization of all operations.

In addition to supporting oscillation, the 33pF capacitor also performs noise filtering by removing unwanted high-frequency disturbances from the signal, thereby improving overall signal clarity and reliability. It contributes to frequency stability, ensuring that the oscillator does not drift or fluctuate due to environmental changes like temperature or electrical noise.

Fig.2.6 capacitor(33pf)

Role-in-Circuit: -

Capacitors are used to provide a stable clock signal and proper reset operation in the 8051 circuit. They help in smooth starting of the microcontroller and reduce noise for stable working. In a password-based door lock system, the 33pF capacitor plays an important role in maintaining the stable operation of the microcontroller's clock circuit. It is used along with the crystal oscillator connected to the microcontroller (such as AT89C51) to provide the required load capacitance, which helps generate a stable and accurate clock signal. This clock signal controls the timing of all system operations, including reading the keypad input, processing the password, updating the LCD display, and controlling the motor driver for locking or unlocking the door. The 33pF capacitor also helps in reducing high-frequency noise and ensuring frequency stability, which prevents errors and ensures reliable system performance. Without it, the oscillator may become unstable, causing improper functioning of the entire door lock system.

3. 8051 Microcontroller

- **Function: -**

  Processes input signals and controls the overall operation of the system. The AT89C51 microcontroller is a powerful 8-bit controller that acts as the central processing unit of an embedded system, performing all control, decision-making, and coordination tasks. It operates by fetching instructions from its internal Flash memory, decoding them, and executing them sequentially to control connected hardware. It continuously monitors input devices such as keypads, switches, or sensors through its four I/O ports (Port 0 to Port 3), processes the received data using its internal CPU and registers, and produces corresponding outputs to devices like LCD displays, LEDs, buzzers, and motor driver ICs. The microcontroller also manages temporary data using its internal RAM and ensures proper execution flow through its program memory.

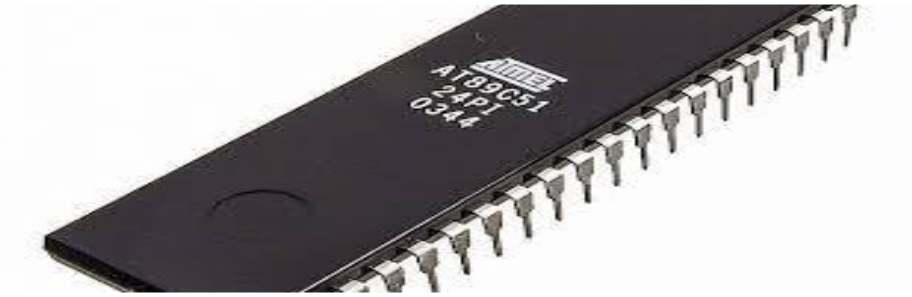
  In addition, the AT89C51 includes timers and counters, which are used to generate accurate delays, count external events, and control timing-based operations such as password entry time limits or motor activation duration. It supports serial communication, allowing it to exchange data with other devices like computers or communication modules. The presence of an interrupt system enables the microcontroller to respond immediately to important external or internal events, improving system efficiency and responsiveness.

  In a password-based door lock system, the AT89C51 plays a critical role by continuously scanning the keypad for user input, comparing the entered password with the stored one, and making decisions based on the result. If the password is correct, it sends signals to a motor driver IC (like L293D) to unlock the door; if incorrect, it may trigger a buzzer or display an error message. At the same time, it updates the LCD display to guide the user through the process. Overall, the AT89C51 ensures smooth, accurate, and automated operation of the entire system by integrating input processing, logical decision-making, timing control, and output management into a single compact unit.

- **Role-in-Circuit: -**

  The 8051 microcontroller acts as the central control unit that manages the entire operation of the password-based door lock system. The AT89C51 microcontroller acts as the brain of the system. It reads the password from the keypad, compares it with the stored password, and controls the outputs like the motor, LCD, and buzzer based on the result. In a password-based door lock system, the AT89C51 microcontroller acts as the central controller (brain) that manages and coordinates all operations of the system. It continuously scans the keypad to detect user input when a password is entered, then stores and processes this input using its internal memory. The entered password is compared with the pre-programmed password stored in the microcontroller. Based on this comparison, the AT89C51 makes a decision—if the password matches, it sends control signals to a motor driver (such as L293D) to unlock the door; if the password is incorrect, it keeps the door locked and may activate a buzzer or display an error message.

At the same time, the microcontroller controls the LCD display to provide instructions and feedback to the user, such as “Enter Password,” “Access Granted,” or “Wrong Password.” It also uses its internal timers to manage delays, such as keeping the door unlocked for a few seconds before locking it again automatically. Additionally, it can handle security features like limiting the number of incorrect attempts or triggering an alarm after multiple failures. Overall, the AT89C51 integrates input handling, decision-making, output control, and timing functions, ensuring the door lock system operates efficiently, securely, and automatically.



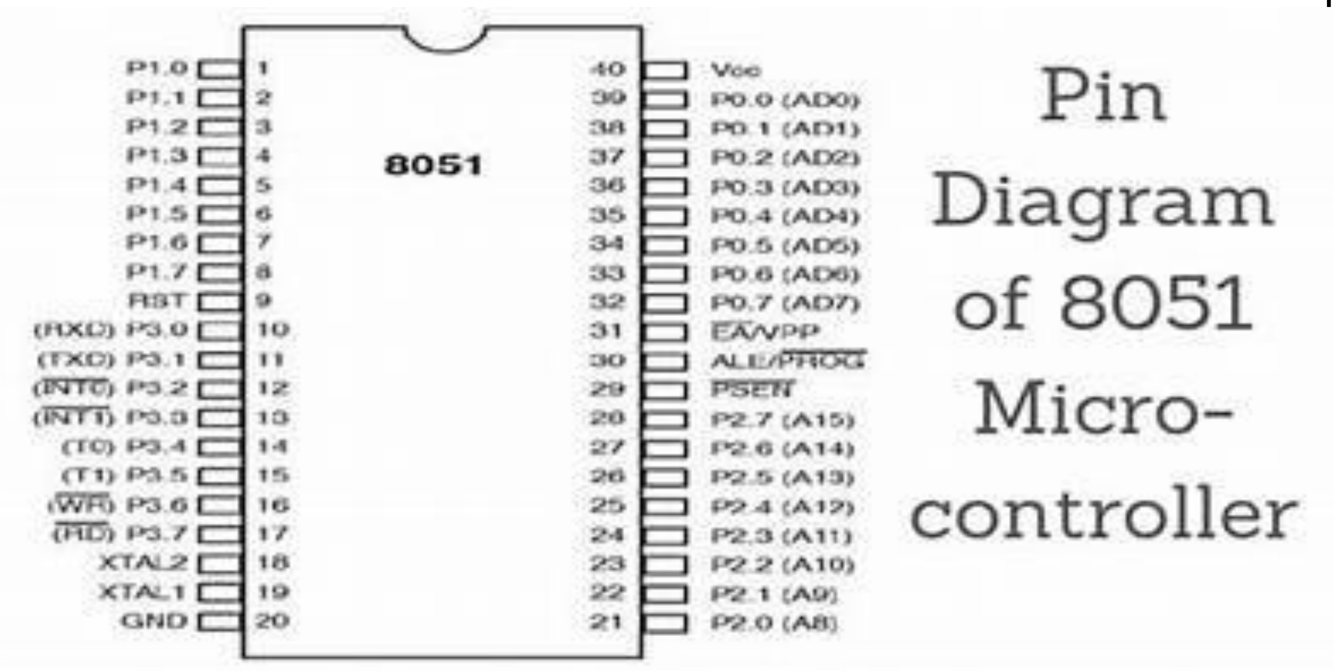


Fig.2.7Microcontroller AT89C51

- **Pin configuration: -**

The AT89C51 is a 40-pin 8-bit microcontroller. Its pins are divided into four ports (P0–P3) and control/system pins. Each pin plays an important role in communication, control, and processing**.** Each pin information is given below.

Pin 1 – P1.0: - General purpose input/output pin of Port 1.

Pin 2 – P1.1: - General purpose input/output pin of Port 1.

Pin 3 – P1.2: - General purpose input/output pin of Port 1.

Pin 4 – P1.3: - General purpose input/output pin of Port 1.

Pin 5 – P1.4: -General purpose input/output pin of Port 1.

Pin 6 – P1.5: -General purpose input/output pin of Port 1.

Pin 7 – P1.6: -General purpose input/output pin of Port 1.

Pin 8 – P1.7: -General purpose input/output pin of Port 1.

Pin 9 – RST: -Reset pin used to restart the microcontroller.

Pin 10 – P3.0 (RXD): -Serial data receiving pin.

Pin 11 – P3.1 (TXD): -Serial data transmitting pin.

Pin 12 – P3.2 (INT0): -External interrupt 0 input pin.

Pin 13 – P3.3 (INT1): -External interrupt 1 input pin.

Pin 14 – P3.4 (T0): -Timer 0 external input pin.

Pin 15 – P3.5 (T1): - external input pin.

Pin 16 – P3.6 (WR): -External memory write control pin.

Pin 17 – P3.7 (RD): -External memory read control pin.

Pin 18 – XTAL2: -Output pin for crystal oscillator.

Pin 19 – XTAL1: -Input pin for crystal oscillator.

Pin 20 – GND: -Ground pin connected to 0V.

Pin 21 – P2.0 (A8): -Port 2 input/output pin and address line A8.

Pin 22 – P2.1 (A9): -Port 2 input/output pin and address line A9.

Pin 23 – P2.2 (A10): -Port 2 input/output pin and address line A10.

Pin 24 – P2.3 (A11): -Port 2 input/output pin and address line A11.

Pin 25 – P2.4 (A12): -Port 2 input/output pin and address line A12.

Pin 26 – P2.5 (A13): -Port 2 input/output pin and address line A13.

Pin 27 – P2.6 (A14): -Port 2 input/output pin and address line A14.

Pin 28 – P2.7 (A15): -Port 2 input/output pin and address line A15.

Pin 29 – PSEN: -Program Store Enable pin used for external program memory access.

Pin 30 – ALE/PROG: -Address Latch Enable pin and programming control pin.

Pin 31 – EA/VPP: - External Access pin and programming voltage pin.

Pin 32 – P0.7 (AD7): -: Port 0 input/output pin and multiplexed address/data line AD7.

Pin 33 – P0.6 (AD6): -Port 0 input/output pin and multiplexed address/data line AD6.

Pin 34 – P0.5 (AD5): -Port 0 input/output pin and multiplexed address/data line AD5.

Pin 35 – P0.4 (AD4): -Port 0 input/output pin and multiplexed address/data line AD4.

Pin 36 – P0.3 (AD3): -Port 0 input/output pin and multiplexed address/data line AD3.

Pin 37 – P0.2 (AD2): -Port 0 input/output pin and multiplexed address/data line AD2.

Pin 38 – P0.1 (AD1): -Port 0 input/output pin and multiplexed address/data line AD1.

Pin 39 – P0.0 (AD0): -Port 0 input/output pin and multiplexed address/data line AD0.

Pin 40 – VCC: -Positive power supply pin (+5V).

- Port 0 (Pins 32–39): –

Port 0 of the AT89C51 microcontroller consists of 8 pins, i.e., P0.0 to P0.7. It is used as general purpose input/output pins as well as a multiplexed address and data bus (AD0–AD7) when interfacing with external memory. This port does not have internal pull-up resistors, so external pull-up resistors are required for proper operation. It performs dual functions, making it important for memory interfacing and data transfer. Port 0 is slightly complex compared to other ports but is very useful in advanced applications.

- Port 1 (Pins 1–8): –

Port 1 of the AT89C51 microcontroller consists of 8 pins, i.e., P1.0 to P1.7. It is used as general purpose input/output pins for interfacing external devices. This port has internal pull-up resistors, so no external resistors are required. It does not have any alternate or special functions, which makes it simple and easy to use. Port 1 is commonly used in applications like keypad interfacing, LCD connections, and controlling other peripherals. It is easy to program and provides reliable operation in embedded systems.

- Port 2 (Pins 21–28): –

Port 2 of the AT89C51 microcontroller consists of 8 pins, i.e., P2.0 to P2.7. It is used as general purpose input/output pins as well as for higher-order address bus (A8–A15) when interfacing with external memory. This port has internal pull-up resistors, so no external resistors are required. It performs dual functions, making it useful in both normal I/O operations and memory interfacing. Port 2 is easy to use and widely used for connecting devices like LCD or other peripherals. It is simple to program and provides efficient operation in embedded systems.

- Port 3 (Pins 10–17): –

Port 3 of the AT89C51 microcontroller consists of 8 pins, i.e., P3.0 to P3.7. It is used as general purpose input/output pins along with special functions. This port has internal pull-up resistors, so no external resistors are required. Each pin has an alternate function such as serial communication (RXD, TXD), external interrupts (INT0, INT1), timer inputs (T0, T1), and control signals (RD, WR) for external memory. Because of these special features, Port 3 is very important and widely used in advanced applications. It is easy to program and useful for communication and control operations in embedded systems.

Pin	**Name**	**Function**
P3.0	RXD	Serial data receiver
P3.1	TXD	Serial data transmitter
P3.2	INT0	External interrupt 0
P3.3	INT1	External interrupt 1
P3.4	T0	Timer 0 input
P3.5	T1	Timer 1 input
P3.6	WR	Write external memory
P3.7	RD	Read from external memory

Table 2.1 Pin Configuration of Port 3

- Power Supply Pins: -

Power supply pins of the AT89C51 microcontroller include two main pins, VCC and GND. These pins are essential for the proper functioning of the microcontroller, as without power supply the system cannot operate. They ensure stable and continuous working of the entire embedded system.

1. VCC (Pin 40) is used to provide the required +5V power supply to the microcontroller for its operation.
2. GND (Pin 20) is connected to ground (0V) and completes the electrical circuit. system.

- Crystal Oscillator pins: -

  Crystal oscillator pins of the AT89C51 microcontroller are XTAL1 (Pin 19) and XTAL2 (Pin 18). These pins are used to connect an external crystal oscillator, which provides the clock signal required for the microcontroller to operate. XTAL1 acts as the input to the internal oscillator circuit, while XTAL2 acts as the output. A crystal (commonly 11.0592 MHz) is connected between these two pins along with capacitors to generate stable clock pulses. These clock signals control the timing and execution of all instructions in the microcontroller, ensuring proper and synchronized operation of the system.

- Reset Pin: -
  Reset pin of the AT89C51 microcontroller is RST (Pin 9). It is an active HIGH pin used to restart the microcontroller. When a high signal is applied to this pin for a short duration (at least two machine cycles), the microcontroller resets and begins execution from the starting address (0000H). It is typically connected with a capacitor and resistor to form a reset circuit, which automatically resets the system when power is turned ON.
- Address Latch Enable Pin: -
  ALE pin of the AT89C51 microcontroller is Pin 30 and stands for Address Latch Enable. It is used to separate (demultiplex) the address and data signals from Port 0 when external memory is interfaced. During operation, Port 0 carries both address and data, so ALE provides a signal to latch the lower 8-bit address into an external latch. This allows the microcontroller to use the same lines for both address and data at different times. ALE is mainly important in systems using external memory and is automatically generated by the microcontroller.
- Program Store Enable Pin: -
  PSEN pin of the AT89C51 microcontroller is Pin 29 and stands for Program Store Enable. It is used to read program code from external memory. This pin is an active LOW signal, meaning it becomes LOW when the microcontroller accesses external program memory. PSEN is mainly used when the program is stored outside the microcontroller, and it enables the external memory to send data to the CPU. It plays an important role in systems where external ROM is used.
- External Access Pin: -
  EA pin of the AT89C51 microcontroller is Pin 31 and stands for External Access. It is used to select whether the microcontroller will use internal program memory or external program memory. If EA is connected to HIGH (VCC), the microcontroller executes the program from its internal memory. If EA is connected to LOW (GND), it accesses program code from external memory. This pin is important in applications where external ROM is required, but in most simple projects it is tied to HIGH to use internal memory.

4. Crystal Oscillator

- **Function:** -

  An 11.0592 MHz crystal oscillator is a key component used in microcontroller systems (especially with the AT89C51) to generate a precise and stable clock signal. This clock signal acts as the heartbeat of the microcontroller, controlling the speed and timing of all internal operations. The crystal works based on the piezoelectric effect, where it vibrates at a fixed frequency (11.0592 MHz) when voltage is applied, producing a highly accurate oscillating signal.

  The main function of this crystal oscillator is to provide timing synchronization for instruction execution, data processing, and communication. In the 8051 microcontroller, one machine cycle typically requires 12 clock pulses, so with an 11.0592 MHz crystal, the microcontroller operates with very precise timing. This specific frequency is widely used because it allows accurate baud rate generation for serial communication (like UART), making it ideal for communication with devices such as computers or GSM modules.

  The crystal is usually connected with two 33pF capacitors, which help in stabilizing the oscillations and ensuring reliable startup of the oscillator circuit. Together, they form a stable clock generation system that ensures the microcontroller performs tasks like reading keypad input, controlling displays, generating delays, and operating actuators with high accuracy.

- **Role-in-Circuit:** -

  The crystal oscillator provides a stable clock signal (typically 11.0592 MHz) to the microcontroller, enabling proper timing and execution of instructions. In a password-based door lock system, the crystal oscillator (commonly 11.0592 MHz) plays a vital role by providing a stable and accurate clock signal to the microcontroller (such as AT89C51). This clock signal acts as the timing reference for all operations performed by the system. Every task—like scanning the keypad for password input, comparing the entered password with the stored one, updating the LCD display, and controlling the motor driver (L293D) to lock or unlock the door—depends on this precise timing.

The crystal oscillator ensures that all instructions inside the microcontroller are executed at the correct speed and in proper sequence. It also helps in generating accurate delays, such as keeping the door unlocked for a fixed time before locking it again, and supports reliable serial communication if the system includes communication modules. Additionally, when used with capacitors (like 33pF), it maintains frequency stability and reduces noise, preventing errors or unpredictable behavior.

Fig.2.8Crystal Oscillator(11.0592mhz)

5. Lcd Display (LM016L)

- **Function: -**

  Lcd display is used to show messages and system status.

- **Role-in-circuit: -**

  The LCD display shows messages like "Enter Password" and "Access Granted." It is connected to the AT89C51 microcontroller and helps the user understand the system status.

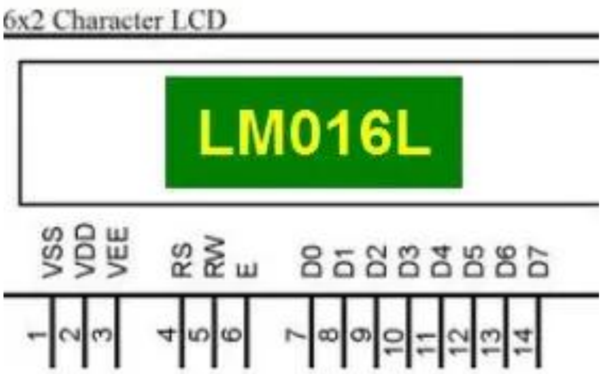


Fig2.9 LCD Display (LM016L)

- **Pin configuration: -**

Pin no.	**Name**	**Function**
1	VSS	Ground voltage
2	VDD	+5v
3	VCC	Contrast voltage
4	RS	Register selects 1-Data register 0-Instruction register
5	R/W	Read/Write mode, to select read/write mode 0-write mode 1-read mode
6	E	Enable 0-start to latch data to LCD character 1-Disable
7	DB0	Data bit 0(LSB BIT)
8	DB1	Data bit 1
9	DB2	Data bit 2
10	DB3	Data bit 3
11	DB4	Data bit 4
12	DB5	Data bit 5
13	DB6	Data bit 6
14	DB7	Data bit 7(MSB)
15	BPL	Black plane light(+5v) or lower (optional)
16	GND	Ground voltage (optional)

Table2.2 Pin Configuration of LCD Display

- **LCD Commands: -**

Command Name	**Hexadecimal Value**	**Description**
Clear display	0x01	Clears the entire display and returns cursor to home position
Return Home	0x02	Moves the cursor to the home position (0, 0)
Entry Mode Set	0x04	Sets cursor movement direction and display shift options
Entry Mode	0x05	Shift display left
Entry Mode	0x06	Sets cursor movement in right
Entry Mode	0x07	Shift display right
Display Off Control	0x08	Controls display, cursor, and cursor blinking options
Display ON	0x0C	Display ON, Cursor Off
Cursor ON	0x0E	Display ON, Cursor ON
Cursor Blink	0x0F	Cursor Blink
Cursor or Display Shift	0x10	Shifts the cursor or the entire display left
Cursor moves	0x14	Cursor moves right
Display shift	0x18	Shift display left
Display shift	0x1C	Shift display right
Function Set	0x20	Sets LCD data length, number of display lines, and font size
Function Set	0x28	4-bit,2line,5x7font
Function Set	0x30	8-bit,1line,5x7font
Function Set	0x38	8-bit,2line,5x7font
Set CGRAM Address	0x40	Sets address of Character Generator RAM (CGRAM) for custom character creation
Set DDRAM Address	0x80	Sets address of Display Data RAM (DDRAM) for writing data to a specific location on the LCD

Table2.3 LCD Commands

6.Motor Driver(l293d)

- **Function: -**

L293D motor driver IC driver the motor by amplifying current. The function of the L293D IC is to act as a motor driver that allows a microcontroller or digital circuit to control DC motors and stepper motors efficiently. Since microcontrollers provide only low current, the L293D performs current amplification, converting low-power input signals into high-power outputs capable of driving motors. It uses an internal dual H-bridge circuit to control the direction of motor rotation, enabling the motor to move forward or reverse by changing the polarity of the voltage applied to it. The IC also supports speed control through techniques like PWM (Pulse Width Modulation), usually applied to the enable pins. Additionally, it can control two motors independently or a single stepper motor due to its dual-channel design. The L293D includes internal protection diodes to prevent damage from back EMF generated by motors and allows separate power supplies for logic and motor operation, ensuring safe and efficient performance. Overall, its main function is to interface, control, and protect motor operations in embedded and robotic systems.

- **Role-in-circuit: -**

The L293D motor driver IC controls the motor by boosting current from the AT89C51 microcontroller enabling door opening and closing.it is a 16 pin IC. In a password-based door lock system, the L293D IC plays a crucial role as a motor driver that controls the locking and unlocking mechanism of the door. The system typically consists of a keypad, microcontroller, LCD display, L293D IC, and a DC motor or solenoid lock. When a user enters a password through the keypad, the microcontroller verifies it. If the password is correct, the microcontroller sends control signals to the L293D IC. Since the microcontroller cannot directly drive a motor due to its low current output, the L293D acts as an interface and current amplifier, enabling the motor to operate. The L293D uses its internal H-bridge circuit to control the direction of rotation of the motor, which is essential for locking and unlocking. For example, when the correct password is entered, the L293D rotates the motor in one direction to open (unlock) the door, and after a delay or another command, it rotates the motor in the opposite direction to close (lock) the door. If the password is incorrect, the L293D does not activate the motor, keeping the door locked. Additionally, the L293D provides protection against back EMF generated by the motor using its internal diodes, ensuring the safety of the circuit. It also allows separate voltage supplies for logic and motor operation, improving system reliability.

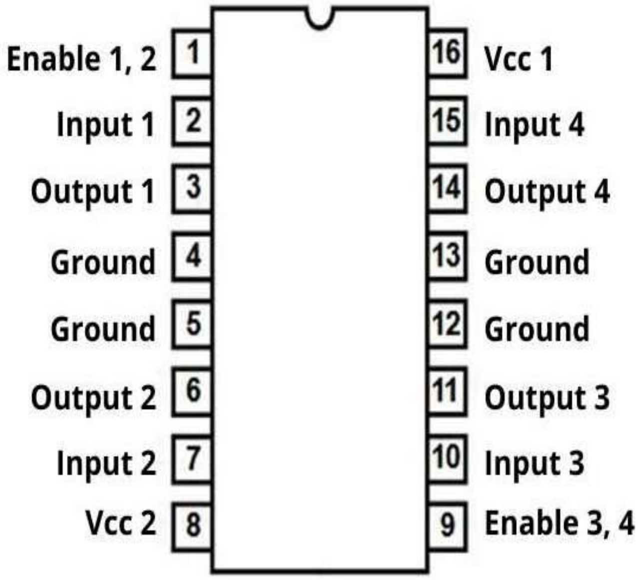


Fig2.10 Motor Driver Circuit(L293D)

Pin no.	**Name**	**Function**
1	Enable 1,2	Enables Motor 1 (must be HIGH to operate)
2	Input 1	Input control for Motor 1
3	Output 1	Connected to Motor 1 terminal
4	GND	Ground
5	GND	Ground
6	Output 2	Connected to Motor 1 termina
7	Input 2	Input control for Motor 1
8	Vcc2	Motor power supply (up to 36V)
9	Enable 3,4	Enables Motor 2
10	Input 3	Input control for Motor 2
11	Output 3	Connected to Motor 2 terminal
12	GND	Ground
13	GND	Ground
14	Output 4	Connected to Motor 2 termina
15	Input 4	Input control for Motor 2
16	Vcc1	Logic power supply (+5V)

Table2.4pin Configuration of L293d Motor Drier

7.Keypad(4x4): -

- **Function: -**

Its function is to allow the user to input the password, and its role in the circuit is to provide data to the microcontroller, which then processes it to grant or deny access. The function of a keypad is to act as an input device that allows the user to enter data, numbers, or commands into a system. It works by detecting which key is pressed using a matrix arrangement of rows and columns, where pressing a key creates a connection between a specific row and column. The keypad performs scanning to identify this position, then uses debouncing to remove noise caused by mechanical switching and ensure accurate input. After detection, it encodes the key into a suitable format such as a number or ASCII value and sends it to the microcontroller through data transfer. Thus, the keypad's overall function is to detect, process, and transmit user input reliably to the system.

- **Role-in-circuit: -**

The 4×4 keypad is used as an input device to enter the password in the system. It consists of 16 keys arranged in rows and columns, and each key press sends a signal to the AT89C51 microcontroller.

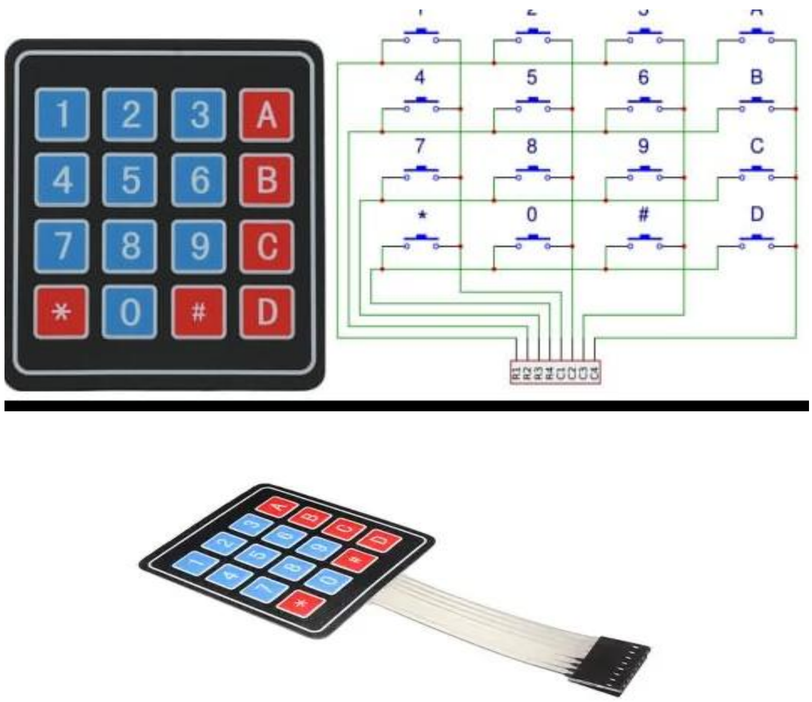


Fig2.11 Keypad(4x4)

8.Motor: -

- **Function: -**

A motor is an electrical device that converts electrical energy into mechanical energy. It produces rotational motion, which is used to perform physical work such as moving, rotating, or driving mechanical systems.

- **Role-in-circuit: -**

The motor is used to open and close the door lock mechanism. It receives control signals through the L293D motor driver IC, which is driven by the AT89C51 microcontroller, allowing the door to unlock or lock automatically.

Fig2.12 Motor

9.Potentiometer(pot 10kohm):-

- **Function:-**

A potentiometer (pot) is a variable resistor used to adjust voltage levels in a circuit. The function of a 10k potentiometer is to provide adjustable control of voltage and resistance in an electronic circuit. It is a three-terminal device consisting of two fixed ends connected to a resistive track and a movable middle terminal called the wiper. When the knob is rotated, the wiper moves along the resistive path, changing the resistance between the terminals and thereby varying the output voltage.

The most important function of a 10k potentiometer is to act as a voltage divider. In this mode, one end is connected to the supply voltage, the other to ground, and the output is taken from the wiper. As the position of the wiper changes, the input voltage is divided in different proportions, producing a variable output voltage between 0V and the maximum supply voltage. This allows smooth and continuous adjustment of voltage levels in a circuit.

- **Role-in-circuit:-**

In this project, it is mainly used to control the contrast of the LCD display, making the text clearer and easier to read. In a password-based door lock system, the 10k potentiometer plays a supporting but important role, mainly in adjusting and stabilizing the circuit operation, especially for the display unit. It is commonly connected to the LCD (16×2) module to control the contrast of the display, ensuring that the entered password and system messages (like "Enter Password", "Access Granted", or "Wrong Password") are clearly visible to the user. By adjusting the potentiometer, the voltage at the LCD's contrast pin is varied, which improves readability under different lighting conditions. In some designs, it may also be used for fine-tuning voltage levels or calibration within the circuit. Although it does not directly control the locking mechanism like the motor driver, the 10k potentiometer ensures proper visual feedback and reliable operation of the system, making user interaction easier and more effective.

Fig.2.13 Potentiometer(Pot 10kohm)

10.Program:-

```
#include<reg51.h>
#include<string.h>

sbit RS = P3^0;
sbit EN = P3^1;
sbit IN1 = P3^2; // Motor Pin 1
sbit IN2 = P3^3; // Motor Pin 2

void delay(unsigned int a) {
    unsigned int i, j;
    for(i=0; i<a; i++)
        for(j=0; j<255; j++);
}

void cmd(char cm) {
    P2 = cm;
    RS = 0;
    EN = 1;
    delay(1);
    EN = 0;
}

void dat(char dt) {
    P2 = dt;
    RS = 1;
    EN = 1;
    delay(1);
    EN = 0;
}

void display(char *lcd) {
    while(*lcd != '\0') {
```

```
        dat(*lcd);
        lcd++;
    }
}

void lcdint() {
    cmd(0x38); cmd(0x0E); cmd(0x01); cmd(0x80);
}

void main() {
    char pass[5] = "6162";
    char pass2[5];
    int i = 0;

    lcdint();
    display("Password-");

    while(1) {
        // Keypad scanning logic
        while(i < 4) {
            P1 = 0xFE; // Row 1
            if(P1 == 0xEE) { pass2[i] = '7'; dat('7'); delay(200); i++; }
            if(P1 == 0xDE) { pass2[i] = '8'; dat('8'); delay(200); i++; }
            if(P1 == 0xBE) { pass2[i] = '9'; dat('9'); delay(200); i++; }

            P1 = 0xFD; // Row 2
            if(P1 == 0xED) { pass2[i] = '4'; dat('4'); delay(200); i++; }
            if(P1 == 0xDD) { pass2[i] = '5'; dat('5'); delay(200); i++; }
            if(P1 == 0xBD) { pass2[i] = '6'; dat('6'); delay(200); i++; }

            P1 = 0xFB; // Row 3
            if(P1 == 0xEB) { pass2[i] = '1'; dat('1'); delay(200); i++; }
            if(P1 == 0xDB) { pass2[i] = '2'; dat('2'); delay(200); i++; }
            if(P1 == 0xBB) { pass2[i] = '3'; dat('3'); delay(200); i++; }
```

```
        P1 = 0xF7; // Row 4
        if(P1 == 0xD7) { pass2[i] = '0'; dat('0'); delay(200); i++; }
      }

      if(i == 4) {
        pass2[4] = '\0'; // Null terminate the input
        if(strcmp(pass, pass2) == 0) {
          cmd(0xC0);
          display("Correct  ");

          // MOTOR ON
          IN1 = 1; IN2 = 0;
          delay(2000); // Wait 2 seconds

          // MOTOR STOP (Crucial fix)
          IN1 = 0; IN2 = 0;
        } else {
          cmd(0xC0);
          display("Incorrect");
          delay(1000);
        }

        // RESET SYSTEM
        i = 0;
        cmd(0x01);
        display("Password-");
      }
    }
}
```

3.CONCLUSION

3.1 Conclusion

The password-based door lock system using the AT89C51 microcontroller is an effective and reliable solution for modern security needs. This system successfully replaces traditional key-based locks with a digital mechanism, thereby eliminating problems such as loss or duplication of keys. By allowing access only when the correct password is entered, it ensures that unauthorized users are restricted, which enhances overall security.

The integration of components like the keypad, LCD display, buzzer, and motor driver makes the system interactive and user-friendly. The keypad enables easy password entry, the LCD provides clear instructions and status messages, and the buzzer alerts in case of incorrect attempts. The motor driver controls the door mechanism efficiently, ensuring smooth opening and closing operations.

This project also demonstrates the practical application of embedded systems by combining both hardware and software. It helps in understanding microcontroller programming, interfacing techniques, and real-time system design. Moreover, the system is cost-effective, easy to implement, and suitable for various applications such as homes, offices, and lockers.

In conclusion, the project achieves its objective of providing a secure, convenient, and efficient access control system, and it can be further enhanced with advanced features like biometric authentication or IoT-based remote access in the future.

3.2Future Scope

The password-based door lock system developed using the AT89C51 microcontroller can be significantly enhanced in the future to improve security, flexibility, and user convenience. While the current system provides basic password protection, modern security demands more advanced and intelligent solutions.

One major improvement can be the integration of biometric authentication such as fingerprint or face recognition. These methods are more secure because they rely on unique physical characteristics of a person, making unauthorized access almost impossible. This would eliminate the risk of password guessing or sharing.

Another important enhancement is the use of wireless communication technologies like Bluetooth, Wi-Fi, or GSM. By connecting the system to a smartphone, users can lock or unlock the door remotely. With IoT (Internet of Things) integration, the system can send real-time notifications or alerts to the user in case of wrong password attempts or suspicious activity.

The system can also be upgraded by implementing OTP (One-Time Password) based access, where a temporary password is generated for each login attempt. This adds an extra layer of security and is especially useful in high-security areas.

Additionally, features like RFID card access can be included, allowing users to unlock the door using a smart card instead of entering a password. Voice-controlled systems can also be implemented for hands-free operation.

From a functional point of view, the system can be improved by adding automatic door closing mechanisms, time-based access control, and support for multiple users with different passwords. This makes the system suitable for offices, industries, and large organizations.

Overall, the future scope of this project lies in transforming it into a smart, connected, and highly secure access control system by combining embedded systems with modern technologies like IoT, biometrics, and wireless communication.

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